

AMENDMENT NO. 1 OCTOBER 2023

TO

**IS 1885 (Part 712) : 2021/IEC 60050-712 : 1992 ELECTROTECHNICAL
VOCABULARY**

PART 712 ANTENNAS

This Amendment No. 1 is identical with Amendment No.1 of 'IEC 60050-712 : 1992 International Electrotechnical Vocabulary (IEV) — Part 712: Antennas' published in 2021 issued by the International Electrotechnical Commission (IEC).

Price Group 2

Amendment No. 1 to IS 1885 (Part 712) : 2021/IEC 60050-712 : 1992

Part 712

Replace IEV 712-02-33, IEV 712-02-34, IEV 712-02-43, IEV 712-02-51, IEV 712-02-52, IEV 712-02-53, IEV 712-02-55, IEV 712-02-56, IEV 712-03-13, IEV 712-04-59, IEV 712-05-82, IEV 712-06-03, IEV 712-06-11 and IEV 712-06-22 by the following:

712-02-33

beamwidth, <of an antenna>

in a specified plane containing the direction of maximum [radiation](#) or the axis of symmetry of a beam or a [radiation lobe](#), [angle](#) between two directions (corresponding, for example, to a given fraction of the maximum radiation or to the first minimums) on both sides of this direction or axis

Note 1 to entry: The most generally used fraction is half-power beamwidth.

Note 2 to entry: The term "beamwidth" has different meanings in optical fibre communication (see [IEV 731-01-35](#)) and the field of ultrasonics (see [IEV 802-01-20](#)).

712-02-34

electrical boresight

axis of an [antenna](#) determined by its [radiation](#) properties

EXAMPLE The null direction of a conical scanning or monopulse [antenna system](#), the [beam axis](#) of a highly [directional antenna](#).

712-02-43

absolute gain, <of an antenna>

isotropic gain, <of an antenna>

ratio, generally expressed in [decibels](#), of the [radiation intensity](#) produced in a given direction by an [antenna](#) to the radiation intensity that would be obtained if the power accepted by the antenna were radiated equally in all directions

Note 1 to entry: If no direction is specified, the direction of maximum radiation intensity from the given antenna is implied.

Note 2 to entry: When the absolute gain is expressed in decibels, the notation dBi is recommended by the International Telecommunication Union as a unit symbol.

Note 3 to entry: The radiation intensity from an antenna in a given direction is the [power](#) radiated per unit [solid angle](#).

Note 4 to entry: If the antenna is lossless its absolute gain is equal to its [directivity](#) in the same direction.

712-02-51

equivalent isotropically radiated power

EIRP

product of the power supplied by a [radio transmitter](#) to an [antenna](#) and the [absolute gain](#) of the antenna in a given direction

712-02-52

effective radiated power, <with respect to a half-wave dipole>

ERP, <with respect to a half-wave dipole>

product of the [power](#) supplied by a [radio transmitter](#) to an [antenna](#) and the [relative gain](#) of the antenna in a given direction, with respect to a [half-wave dipole](#)

Note 1 to entry: The use of the quantity [equivalent isotropically radiated power](#) (referring to absolute gain) is preferable to that of the quantity effective radiated power (referring to relative gain).

Note 2 to entry: The term "effective radiated power" is also used when the reference antenna is not a half-wave dipole (see [IEV 161-04-16](#)).

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712-02-53

**effective monopole radiated power
EMRP**

product of the [power](#) supplied by a [radio transmitter](#) to an [antenna](#) and the [gain](#) of this antenna relative to a short vertical antenna in a given direction

712-02-55

G/T

figure of merit, <of an antenna>

quotient, usually expressed in logarithmic units, of the absolute gain *G* of an antenna to its [noise temperature](#) *T* referred to the antenna terminals at a specified frequency and in specified conditions of installation and operation

Note 1 to entry: If the noise temperature is expressed in kelvins and the gain as a power ratio, the quantity $10 \log G/T$ is expressed in a unit designated by the symbol dB (K⁻¹), sometimes written incorrectly dBK.

Note 2 to entry: In English, the term "figure of merit" has different meanings in relation to antenna receiving systems (see [IEV 712-02-56](#)), transducers (see [IEV 431-02-15](#)) and piezoelectric and dielectric resonators (see [IEV 561-01-26](#)).

Note 3 to entry: In French, the term "facteur de qualité" has a different meaning for a two-terminal element or circuit (see [IEV 151-15-45](#)), for a resonant circuit (see [IEV 151-15-46](#)), for radiation protection purposes (see [IEV 395-01-39](#)), in relation to antenna receiving systems (see [IEV 712-02-56](#)), and in radiology (see [IEV 881-14-03](#)).

712-02-56

G/T

figure of merit, <of an antenna receiving system>

quotient, usually expressed in logarithmic units, of the absolute gain *G* of an antenna by the antenna-receiver [noise temperature](#) *T* referred to the antenna terminals at a specified frequency and in specified conditions of installation and operation

Note 1 to entry: If the noise temperature is expressed in kelvins and the gain as a power ratio, the quantity $10 \log G/T$ is expressed in a unit designated by the symbol dB (K⁻¹), sometimes written incorrectly dBK.

Note 2 to entry: In English, the term "figure of merit" has different meanings in the fields of antennae in general (see [IEV 712-02-55](#)), transducers (see [IEV 431-02-15](#)) and piezoelectric and dielectric resonators (see [IEV 561-01-26](#)).

Note 3 to entry: In French, the term "facteur de qualité" has a different meaning for a two-terminal element or circuit (see [IEV 151-15-45](#)), for a resonant circuit (see [IEV 151-15-46](#)), for radiation protection purposes (see [IEV 395-01-39](#)), in the field of antennae in general (see [IEV 712-02-55](#)), and in radiology (see [IEV 881-14-03](#)).

712-03-13

contoured-beam antenna

[shaped-beam antenna](#) designed in such a way that, when its beam intersects a given surface, the lines of equal [power flux density](#) incident upon the surface form specified contours

Note 1 to entry: For a satellite antenna, the area on the surface of the Earth within which the antenna gain is at least equal to a specified value, usually 3 dB, relative to its maximum gain is called "footprint of the beam" (see [IEV 725-13-04](#)).

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712-04-59

counterpoise, <of an antenna>

system of [conductors](#), elevated above and capacitively coupled to the [local earth](#), to provide the conductive surface for a vertical [monopole antenna](#)

712-05-82

artificial dielectric, <for radio waves>

[inhomogeneous](#) medium consisting of a regular arrangement of scatterers, usually metallic, embedded in a low-[absolute permittivity dielectric medium](#) and which reacts to [radio waves](#) in a certain [frequency range](#) like a [homogeneous](#) dielectric with a permittivity differing from that of the low-permittivity dielectric

712-06-03

balun

See [IEV 161-04-34](#)

712-06-11

power divider, <of an antenna>

junction box, <of an antenna>

device with one input and two or more outputs, which produces the desired distribution of power from the [feed line](#) to the driven elements of an [antenna](#)

Note 1 to entry: The term "junction box" is generally used when a simple power divider is enclosed in a weatherproof housing; see [IEV 442-08-03](#).

712-06-22

antenna simulator

DEPRECATED: dummy antenna

non-radiating [dissipative](#) network simulating the [input impedance](#) of an [antenna](#) over a specified [frequency range](#) and used as a [load](#) for testing a transmitter